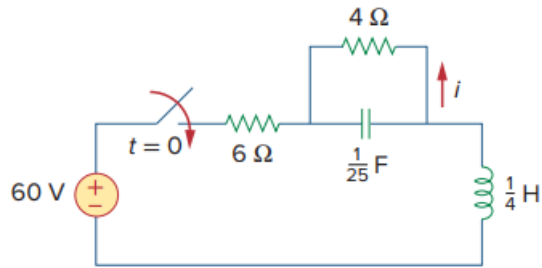


Problem

In the circuit of Fig. 8.102, find $i(t)$ for $t > 0$.

Figure 8.102:



Step-by-step solution

Step 1 of 11

Refer to circuit diagram in Figure 8.102 in the text book.

At $t = 0^-$, switch in the circuit is open, no source is connected to the circuit. Hence, the initial voltage across the capacitor and initial current through the inductor are zero.

$$v_C(0^-) = 0 \text{ V}$$

$$i_L(0^-) = 0 \text{ A}$$

It is known that, the voltage across the capacitor doesn't change instantly.

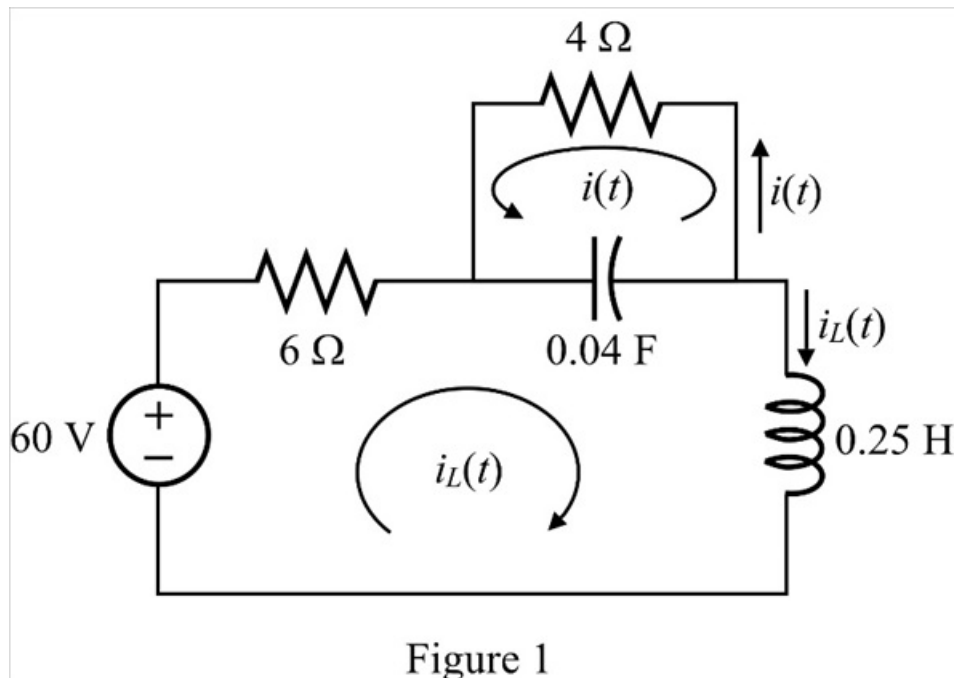
$$\begin{aligned} v_C(0^+) &= v_C(0^-) \\ &= 0 \text{ V} \end{aligned}$$

The current through inductor doesn't change instantly.

$$\begin{aligned} i_L(0^+) &= i_L(0^-) \\ &= 0 \text{ A} \end{aligned}$$

Step 2 of 11

The switch closes at $t = 0$. Draw the circuit diagram at $t > 0$.



Step 3 of 11

Apply Kirchhoff's Voltage Law to loop $i(t)$.

$$4i(t) + \frac{1}{0.04} \int (i(t) + i_L(t)) dt = 0$$

$$25 \int (i(t) + i_L(t)) dt = -4i(t)$$

$$\int (i(t) + i_L(t)) dt = -\frac{4}{25} i(t)$$

Differentiate the equation with respect to t .

$$i(t) + i_L(t) = -\frac{4}{25} \frac{di(t)}{dt}$$

$$i_L(t) = -\frac{4}{25} \frac{di(t)}{dt} - i(t) \dots\dots (1)$$

Step 4 of 11

Apply Kirchhoff's Voltage Law to loop $i_L(t)$.

$$6i_L(t) + \frac{1}{0.04} \int [i_L(t) + i(t)] dt + 0.25 \frac{di_L(t)}{dt} = 60$$

Substitute $-\frac{4}{25}i(t)$ for $\int (i(t) + i_L(t)) dt$.

$$6i_L(t) - \left(\frac{1}{0.04}\right) \left(\frac{4}{25}i(t)\right) + 0.25 \frac{di_L(t)}{dt} = 60$$

Substitute $-\frac{4}{25} \frac{di(t)}{dt} - i(t)$ for $i_L(t)$.

$$6 \left[-\frac{4}{25} \frac{di(t)}{dt} - i(t) \right] - 4i(t) + 0.25 \frac{d}{dt} \left[-\frac{4}{25} \frac{di(t)}{dt} - i(t) \right] = 60$$

$$-\frac{24}{25} \frac{di(t)}{dt} - 6i(t) - 4i(t) - \frac{1}{25} \frac{d^2i(t)}{dt^2} - \frac{1}{4} \frac{di(t)}{dt} = 60$$

$$-\frac{1}{25} \frac{d^2i(t)}{dt^2} - 1.21 \frac{di(t)}{dt} - 10i(t) = 60$$

$$\frac{d^2i(t)}{dt^2} + 30.25 \frac{di(t)}{dt} + 250i(t) = -1500$$

Step 5 of 11

Consider the characteristic equation of the circuit.

$$(s^2 + 30.25s + 250)i = 0$$

$$(s + 15.125 - j4.608)(s + 15.125 + j4.608) = 0$$

The roots of the characteristic equation are,

$$s = -15.125 \pm j4.608$$

The circuit exhibits under-damped response since the roots of the characteristic equation are complex.

Compare the roots to $s = -\alpha \pm j\omega_d$.

The values of α and ω_d are 15.125 Np/s and 4.608 rad/s respectively.

Write the general solution for the current $i(t)$ for under-damped condition.

$$i(t) = i(\infty) + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

Substitute 15.125 Np/s for α and 4.608 rad/s for ω_d .

$$i(t) = i(\infty) + (A_1 \cos 4.608t + A_2 \sin 4.608t) e^{-15.125t} \dots\dots (2)$$

Step 6 of 11

At $t = \infty$ switch in the circuit is closed, inductor acts as short circuit and capacitor acts as open circuit. Draw the circuit diagram for $t = \infty$.

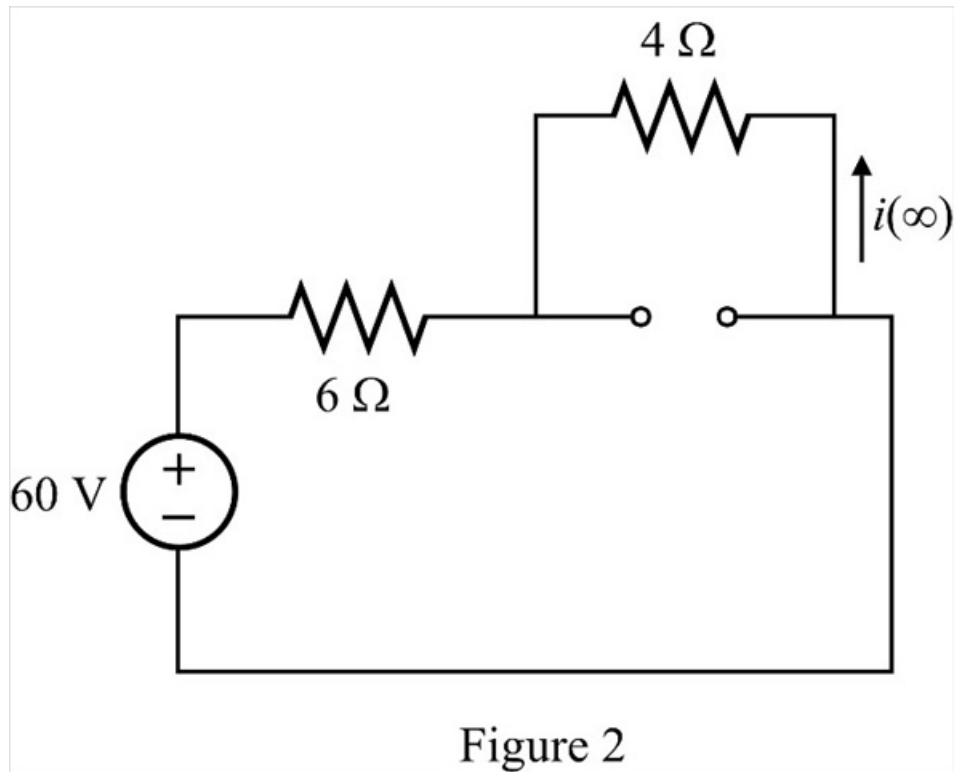


Figure 2

Step 7 of 11

Calculate the current $i(\infty)$.

$$\begin{aligned} i(\infty) &= \frac{-60 \text{ V}}{6 \Omega + 4 \Omega} \\ &= \frac{-60 \text{ V}}{10 \Omega} \\ &= -6 \text{ A} \end{aligned}$$

Recall equation (2).

$$i(t) = i(\infty) + (A_1 \cos 4.608t + A_2 \sin 4.608t)e^{-15.125t}$$

Substitute -6 A for $i(\infty)$.

$$i(t) = -6 + [A_1 \cos(4.608t) + A_2 \sin(4.608t)]e^{-15.125t} \dots\dots (3)$$

Substitute 0 for t .

$$i(0) = -6 + [A_1 \cos(4.608 \times 0) + A_2 \sin(4.608 \times 0)]e^{-15.125(0)}$$

$$0 = -6 + [A_1(1) + A_2(0)](1) \quad [\text{since } i(0) = 0]$$

$$0 = -6 + A_1$$

$$A_1 = 6$$

Step 8 of 11

Recall equation (3).

$$i(t) = -6 + [A_1 \cos(4.608t) + A_2 \sin(4.608t)]e^{-15.125t}$$

Differentiate the equation with respect to t .

$$\frac{di(t)}{dt} = \left\{ \begin{aligned} &0 + (-15.125)[A_1 \cos(4.608t) + A_2 \sin(4.608t)]e^{-15.125t} + \\ &(4.608)[-A_1 \sin(4.608t) + A_2 \cos(4.608t)]e^{-15.125t} \end{aligned} \right\}$$

Substitute 0 for t .

$$\begin{aligned} \frac{di(0)}{dt} &= \left\{ \begin{aligned} &0 + (-15.125)[A_1 \cos(0) + A_2 \sin(0)]e^{-15.125(0)} + \\ &(4.608)[-A_1 \sin(0) + A_2 \cos(0)]e^{-15.125(0)} \end{aligned} \right\} \\ &= (-15.125)[A_1 + A_2(0)] + (4.608)[-A_1(0) + A_2] \\ &= -15.125A_1 + 4.608A_2 \dots\dots (4) \end{aligned}$$

Step 9 of 11

Draw the circuit diagram at $t = 0$.

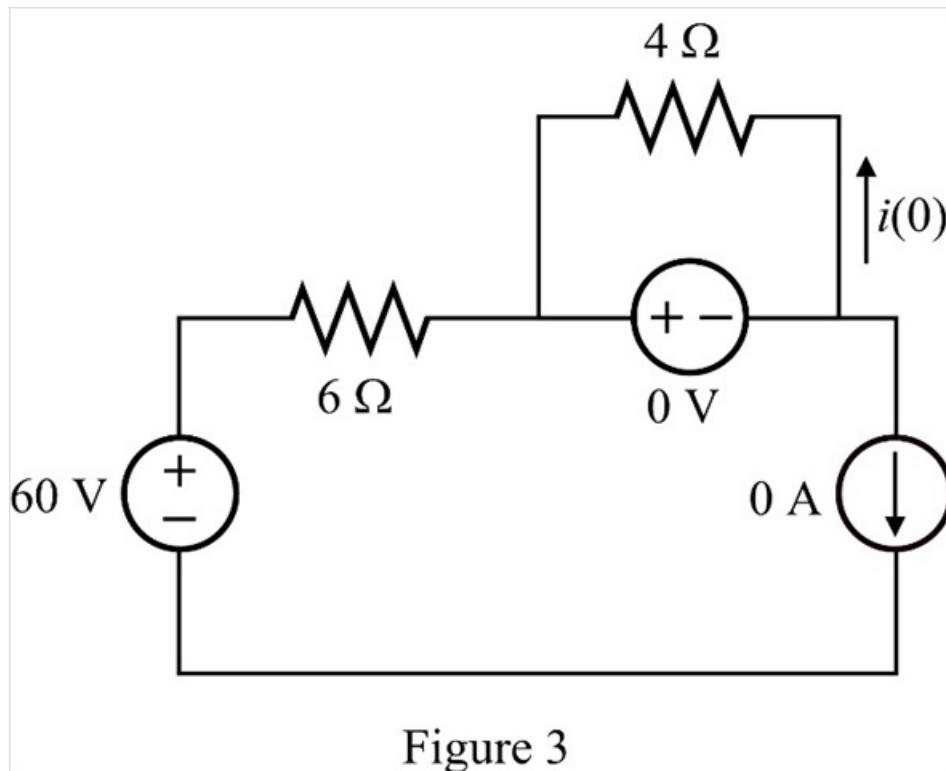


Figure 3

Step 10 of 11

The current $i(0)$ is,

$$i(0) = 0 \text{ A}$$

Differentiate the equation with respect to t .

$$\frac{di(0)}{dt} = 0 \text{ A/s}$$

Recall equation (4).

$$\frac{di(0)}{dt} = -15.125A_1 + 4.608A_2$$

Substitute 0 A/s for $\frac{di(0)}{dt}$ and 6 for A_1 .

$$0 = -(15.125)(6) + 4.608A_2$$

$$4.608A_2 = 90.75$$

$$A_2 = 19.69$$

Step 11 of 11

The current through the inductor for $t > 0$ is,

Recall equation (3).

$$i(t) = -6 + [A_1 \cos(4.608t) + A_2 \sin(4.608t)]e^{-15.125t}$$

Substitute 6 for A_1 and 19.69 for A_2 .

$$\begin{aligned} i(t) &= -6 + [6 \cos(4.608t) + 19.69 \sin(4.608t)]e^{-15.125t} \text{ A} \\ &= [6 \cos(4.608t) + 19.69 \sin(4.608t)]e^{-15.125t} - 6 \text{ A} \end{aligned}$$

Therefore, the current $i(t)$ for $t > 0$ is $\boxed{[6 \cos(4.608t) + 19.69 \sin(4.608t)]e^{-15.125t} - 6 \text{ A}}$.